

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger borrower bases, increased financial data loads, and advanced AI-powered debt management features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Dependence on Internet Connectivity	Users cannot access AI-powered settlement recommendations and negotiation services without an internet connection.	High
2	Gemini API Usage Limits	High volumes of AI requests may affect response availability and increase processing time.	Medium
3	Limited Financial Data Integration	The platform currently supports manual loan data entry and does not integrate directly with banking or financial institutions.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of concurrent borrowers.	Upgrade cloud hosting resources and implement load balancing to support thousands of users simultaneously.
2	Data Storage	Uses SQLite for financial data storage.	Migrate to scalable cloud databases such as PostgreSQL or Google Cloud SQL for improved performance and reliability.
3	AI Performance	AI recommendations depend on Google Gemini API response	Implement response caching, asynchronous processing, and backend optimization to improve AI performance.

		time.	
4	Security	Basic user authentication and API key protection.	Implement multi-factor authentication (MFA), role-based access control, data encryption, and enhanced security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User Authentication, Profile Management & Secure Financial Dashboard	3 Months	Improved security, personalized borrower experience, and secure access to financial information.
Phase 3	Bank API Integration, Credit Score Analysis & Automated EMI Tracking	6 Months	Real-time financial insights, automated loan monitoring, and better debt management decisions.
Phase 4	Mobile Application, Multi-language Support & Predictive Financial Analytics	9 Months	Increased accessibility, wider user adoption, and intelligent financial planning with AI-driven predictions.